

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Enterprise Java (EJ)

Practical – I

Roll No.: T016	Name: Himani Malankar
Class: TYIT	Batch: 1
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AIM: Implement the following Simple Servlet applications.

A. Implement the following Simple Servlet applications.

CODE: (index.html)

```
<!DOCTYPE html>
<html>
<head>
  <title>Calculator</title>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>

  <div>Calculator</div>

  <form action="Calculator" name="frm">

    <label for="num1">Enter Number 1:</label>
    <input name="num1" type="text" />

    <br><br>

    <label for="num2">Enter Number 2:</label>
    <input name="num2" type="text" />

    <br><br>

    <label for="operator">Operator:</label>
    <select name="operator">
      <option value="Addition">+</option>
      <option value="Subtraction">-</option>
      <option value="Multiplication">*</option>
      <option value="Division">/</option>
    </select>

    <br><br>

    <input type="submit" value="Submit" />

  </form>

</body>
</html>
```

SERVLET: (Calculator.java)

```
import java.io.IOException;
import java.io.PrintWriter;
```

```

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet(urlPatterns = {"/Calculator"})
public class Calculator extends HttpServlet {

    @Override
    public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        PrintWriter out = response.getWriter();

        String num1 = request.getParameter("num1");
        String num2 = request.getParameter("num2");
        String op = request.getParameter("operator");

        if (op.equals("Addition")) {
            out.println("Addition is: "
                + (Integer.parseInt(num1) + Integer.parseInt(num2)));
        } else if (op.equals("Subtraction")) {
            out.println("Subtraction is: "
                + (Integer.parseInt(num1) - Integer.parseInt(num2)));
        } else if (op.equals("Multiplication")) {
            out.println("Multiplication is: "
                + (Integer.parseInt(num1) * Integer.parseInt(num2)));
        } else {
            out.println("Division is: "
                + (Integer.parseInt(num1) / Integer.parseInt(num2)));
        }
    }
}

```

OUTPUT:

Calculator

Enter Number 1:

Enter Number 2:

Operator:

Calculator

Enter Number 1:

Enter Number 2:

Operator:

Calculator

Enter Number 1:

Enter Number 2:

Operator:

Calculator

Enter Number 1:

Enter Number 2:

Operator:

Addition is: 44

Subtraction is: -22

Multiplication is: 363

Division is: 0

B. Create a servlet for a login page. If the username and password are correct then it says message “Hello <username>” else a message “login failed”.

CODE:(index.html)

```
<!DOCTYPE html>
<html>
<head>
  <title>Login</title>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>

  <form action="login" method="get" name="frm">

    <label for="uname">Enter Username:</label>
    <input name="uname" type="text" />

    <br><br>

    <label for="pass">Enter Password:</label>
    <input name="pass" type="password" />

    <br><br>

    <input name="submit" type="submit" value="Submit" />
    <input name="reset" type="reset" value="Reset" />

  </form>

</body>
</html>
```

SERVLET: (login.java) Correct usname & pass: user & pwd

```
import java.io.IOException;
import java.io.PrintWriter;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet(urlPatterns = {"/login"})
public class login extends HttpServlet {

    @Override
    public void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");

        try (PrintWriter out = response.getWriter()) {

            String uname = request.getParameter("uname");
            String pass = request.getParameter("pass");

            if (uname.equals("user") && pass.equals("pwd")) {
                out.println("Hello <b>" + uname
                    + "</b> Welcome to Java Servlet");
            } else {
                out.println("Login Failed");
            }
        }
    }
}
```

```
} }
```

OUTPUT:

Enter Username: user Enter Password: ... Submit Reset	Hello user Welcome to Java Servlet
Enter Username: abc Enter Password: ... Submit Reset	Login Failed

C. Create a registration servlet in Java using JDBC. Accept the details such as Username, Password, Email, and Country from the user using HTML Form and store the registration details in the database.

CODE:(index.html)

```
<!DOCTYPE html>
<html>
<head>
  <title>Practical 1 C</title>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>

<body>

  <form action="ServReg" method="post">

    <table cellpadding="5">

      <thead>
        <tr>
          <th colspan="2">REGISTRATION FORM</th>
        </tr>
      </thead>

      <tbody>

        <tr>
          <td><b>First Name:</b></td>
          <td><input type="text" name="fname"></td>
        </tr>

        <tr>
          <td><b>Last Name:</b></td>
          <td><input type="text" name="sname"></td>
        </tr>

        <tr>
          <td><b>Zip Code:</b></td>
          <td><input type="text" name="zip"></td>
        </tr>

        <tr>
          <td><b>User Name:</b></td>
          <td><input type="text" name="uid"></td>
        </tr>

        <tr>
          <td><b>Password:</b></td>
          <td><input type="password" name="pwd"></td>
        </tr>

        <tr>
          <td><b>Confirm Password:</b></td>
          <td><input type="password" name="pwd1"></td>
        </tr>

        <tr>
          <td><b>Town:</b></td>
          <td><input type="text" name="town"></td>
        </tr>

        <tr>
          <td><b>Country:</b></td>
          <td><input type="text" name="country"></td>
        </tr>

      </tbody>
    </table>
  </form>
</body>
</html>
```

```

        <tr>
            <td colspan="2">
                <input type="submit" value="Submit">
            </td>
        </tr>

    </tbody>

</table>

</form>

</body>
</html>

```

SERVLET:

```

import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.SQLException;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet(urlPatterns = {"/ServReg"})
public class ServReg extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");

        try (PrintWriter out = response.getWriter()) {

            String connectionURL = "jdbc:mysql://localhost:3306/db";
            Connection connection = null;

            String uid = request.getParameter("uid");
            String fname = request.getParameter("fname");
            String sname = request.getParameter("sname");
            String pwd = request.getParameter("pwd");
            String pwd1 = request.getParameter("pwd1");
            String town = request.getParameter("town");
            String country = request.getParameter("country");
            String zip = request.getParameter("zip");

            try {

                Class.forName("com.mysql.cj.jdbc.Driver");

                connection = DriverManager.getConnection(
                    connectionURL,
                    "root",
                    "mysql_2026"
                );

                String sql = "INSERT INTO user_register VALUES(?,?,?,?,?,?,?)";

                PreparedStatement pst = connection.prepareStatement(sql);

                pst.setString(1, uid);
                pst.setString(2, fname);

```

```

        pst.setString(3, sname);
        pst.setString(4, pwd);
        pst.setString(5, pwd1);
        pst.setString(6, town);
        pst.setString(7, country);
        pst.setString(8, zip);

        int numRowsChanged = pst.executeUpdate();

        out.println("<h2>Welcome : " + fname + "</h2>");
        out.println("<p>Rows Inserted : " + numRowsChanged + "</p>");

        pst.close();
    } catch (ClassNotFoundException e) {
        out.println("Couldn't Load Database Driver : " + e.getMessage());
    } catch (SQLException e) {
        out.println("SQL Exception Caught : " + e.getMessage());
    } catch (Exception e) {
        out.println(e);
    } finally {
        try {
            if (connection != null) {
                connection.close();
            }
        } catch (SQLException ignored) {
            out.println(ignored);
        }
    }
}
}
}
}
}

```

DATABASE CODE:

```

CREATE DATABASE db;

USE db;

CREATE TABLE user_register (
    uid CHAR(40),
    fname CHAR(40),
    sname CHAR(40),
    pwd CHAR(40),
    pwd1 CHAR(40),
    town CHAR(40),
    country CHAR(40),
    zipcode CHAR(40)
);

```

OUTPUT:

REGISTRATION FORM

First Name:	Sakshi
Last Name:	Malankar
Zip Code:	123456
User Name:	sakshi_02
Password:	
Confirm Password:	
Town:	Mumbai
Country:	India
Submit	

Welcome : Sakshi

Rows Inserted : 1

```
[mysql> select * from user_register;
```

uid	fname	sname	pwd	pwd1	town	country	zipcode
himani_14	Himani	Malankar	hima_1406	hima_1406	Mumbai	India	24680
shubham_21	Shubham	Mandavkar	shub_21	shubh_21	Mumbai	India	13579
sakshi_02	Sakshi	Malankar	sak_02	sak_02	Mumbai	India	123456

```
3 rows in set (0.006 sec)
```